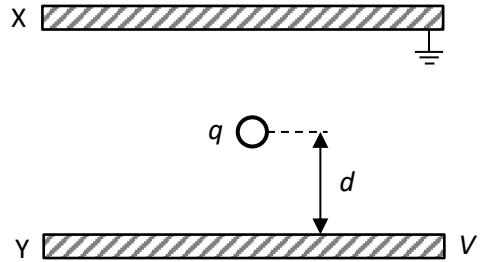
19 An oil drop of charge q and mass m is suspended at the midpoint of parallel charged plates X and Y. The drop is at a distance d from plate Y. Plate X is earthed while Y is held at a potential of V with a total charge of Q_Y.



What is the expression for the potential V?

A $\dfrac{mgd}{q}$

B $\dfrac{2mgd}{q}$

C $\dfrac{mg}{q}$

D $\dfrac{Q_Y}{4\pi\varepsilon_0 d}$